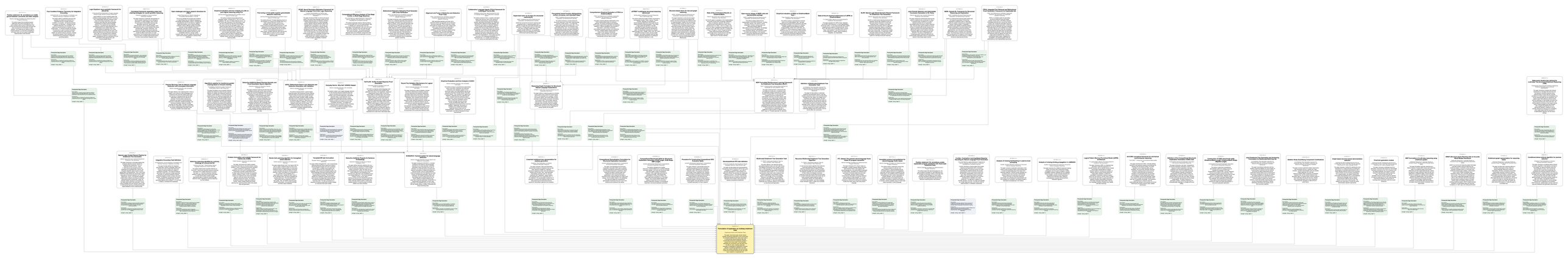

266177015.c4-1

Position statement: NL annotations as middle ground between comments and static analysis

Natural Language Annotations for Reasoning about Program Semantics

The authors argue that annotating program statements with natural-language pre- and post-conditions offers a level of abstraction that is more structured than informal code comments yet does not require the full formalism of static analysis. This middle ground captures intuitive program behavior in a way that is accessible to non-expert users and can support explainable reasoning about code execution.

281421016.c1

Four-Condition Evidence Taxonomy for Integrative Grounding

INTEGROUND: On the Evaluation of Verification and Retrieval Planning in Integrative Grounding

The authors define a systematic taxonomy that classifies the evidence set retrieved for a hypothesis into four mutually exclusive conditions: (i) complete – all required premises are present; (ii) redundant – complete set plus additional distractors; (iii) incomplete – at least one required premise is missing; and (iv) uninformative – no relevant premises are included. This taxonomy is used to construct evaluation instances that probe model behavior under realistic sub-optimal evidence scenarios, enabling fine-grained analysis of both verification and retrieval-planning components in integrative grounding systems.

287365102.c5

Logic-Explainer neuro-symbolic framework for ethical NLI

Enhancing Ethical Explanations of Large Language Models through Iterative Symbolic Refinement

Logic-Explainer is a neuro-symbolic system that combines a large language model (GPT-3.5-turbo) with a differentiable backward-chaining Prolog solver (NLProlog) to generate, verify, and iteratively refine step-wise natural-language explanations for ethical natural-language inference. The pipeline first uses semantic prompting to produce an initial explanation and a hypothesised moral-violation label, then auto-formalises the explanation into Prolog, validates logical correctness with the solver, and feeds back abductive and deductive refinements until a valid, non-redundant proof is obtained. The framework is designed to improve logical validity, completeness and non-redundancy of LLM explanations while aligning them with moral-foundation theory.

268601153.c0

Conceptual framework separating output and chaining strategies for neural symbolic reasoning

Empirical Investigation of Neural Symbolic Reasoning Strategies

The paper introduces a systematic decomposition of neural symbolic reasoning into two orthogonal dimensions: (i) output strategy, which controls the granularity of intermediate reasoning step generation (all-at-once, step-by-step, token-by-token), and (ii) chaining strategy, which determines the order in which equations are solved (shortest-path, exhaustive, backward). This taxonomy clarifies existing neural reasoning approaches and enables controlled experiments that isolate the effect of each factor on model performance. By formalising these dimensions, the authors provide a reusable experimental protocol for future neural-symbolic integration work.

257636734.c6

Open challenges and future research directions for LRNLP

Logical Reasoning over Natural Language as Knowledge Representation: A Survey

The survey identifies the principal open problems in Logical Reasoning over Natural Language as Knowledge Representation (LRNLP), including high computational cost of LLM inference, fragility to logical perturbations, lack of robust automatic evaluation metrics, difficulty handling incomplete premises, limited scalability of inductive reasoning to raw web corpora, and the need for integrated multi-type reasoning pipelines. It also proposes concrete research avenues such as probabilistic inference, stronger verifier modules, larger and more realistic benchmarks, and hybrid neuro-symbolic pipelines to address these challenges.

260464474.c4

Empirical evaluation and error analysis of LLMs on pure logical reasoning using FLD

Learning Deductive Reasoning from Synthetic Corpus based on Formal Logic

The authors evaluate several state-of-the-art language models—including GPT-4, GPT-3.5-Turbo, LongAlpaca-13B, and a T5-based stepwise prover—on the Formal Logic Deduction (FLD) benchmark in both few-shot in-context and fully fine-tuned settings. They report that even the strongest model, GPT-4, solves only about half of the FLD problems, and that correct final answers are often accompanied by incorrect intermediate proof steps. A detailed error analysis identifies systematic failure modes such as logical hallucination, misuse of distractor facts, and failure to faithfully follow the generated proof sequence, highlighting a gap between answer accuracy and true step-wise reasoning ability.

260464474.c5

Fine-tuning on FLD yields superior generalizable deductive reasoning

Learning Deductive Reasoning from Synthetic Corpus based on Formal Logic

The authors fine-tune T5-based stepwise provers on the synthetic FLD corpora and evaluate them on (i) other synthetic deduction corpora (e.g., RuleTaker, AACorpus variants) and (ii) the human-authored EntailmentBank benchmark. In both settings the FLD-trained models achieve higher proof accuracy than baselines that are not trained on FLD, demonstrating that exposure to axiom-based, complete deduction data improves the models' ability to construct correct multi-step proofs and to transfer to real-world logical reasoning tasks.

268857317.c5

GFaiR: Neural Resolution-Refutation Framework for Natural-Language First-Order Logic Reasoning

Towards Generalizable and Faithful Logic Reasoning over Natural Language via Resolution Refutation

The paper introduces GFaiR, an end-to-end neural system that equips large language models with the complete reasoning power of resolution refutation. GFaiR iteratively selects two natural-language clauses, applies a learned resolution step at the language level to synthesize a new clause, and repeats until a contradiction is derived, thereby determining the truth value of a hypothesis. The architecture comprises a T5-based converter that rewrites facts and rules into a Skolem-like clause form, XLNet-based pre- and post-selectors that choose clause pairs, a T5 knowledge composer that implicitly learns the resolution rule, and a verifier trained with a validity-contrastive loss to enforce logical compatibility of selected pairs.

265150086.c3

Computational Efficiency Analysis of One-Stage ConDec vs Multi-Stage Baselines

Are LLMs Rigorous Logical Reasoners? Empowering Natural Language Proof Generation by Stepwise Decoding with Contrastive Learning

The authors empirically measure inference latency on an NVIDIA A800 GPU for the EntailmentBank Task 2 validation set, reporting 294 ms for the single-stage ConDec generator, 430 ms for NLProofS with search, and 544 ms for NLProofS with both search and verification. This analysis quantifies the runtime reduction achieved by eliminating separate search and verification components while preserving or improving proof-generation accuracy.

271921406.c2

Bidirectional Deductive-Abductive Proof Generator with Cross-Verification

FRVA: Fact-Retrieval and Verification Augmented Entailment Tree Generation for Explainable Question Answering

The paper introduces a pair of seq2seq generators built on T5: a deductive generator that constructs intermediate conclusions from retrieved facts in a bottom-up fashion, and an abductive generator that decomposes the hypothesis into sub-goals in a top-down manner. For each generated step, a verifier based on ALBERT evaluates consistency between the forward (deductive) and backward (abductive) directions, selecting only steps that are mutually supported. This bidirectional generation with cross-verification mitigates error accumulation and improves the structural correctness of entailment trees for explainable question answering.

271921406.c4

Alignment and Fusion of Deductive and Abductive Proof Trees

FRVA: Fact-Retrieval and Verification Augmented Entailment Tree Generation for Explainable Question Answering

After generating separate proof trees with a deductive (bottom-up) generator and an abductive (top-down) generator, FRVA constructs a unified proof graph by merging the two trees, assigns each node a score that combines the generator's log-likelihood and the cross-verification confidence, and then extracts the highest-scoring partial graph as the final entailment tree. This fusion leverages complementary evidence from both reasoning directions to produce a coherent, structurally correct explanation for the hypothesis.

281675908.c1

Collaborative Language Agents (COLA) framework for multilingual Text-to-SQL

Multilingual Text-to-SQL: Benchmarking the Limits of Language Models with Collaborative Language Agents

The authors introduce COLA, a modular multi-agent system that improves Text-to-SQL parsing by iteratively decomposing a natural-language question, generating intermediate SQL fragments, executing them against the target database, and refining the query based on execution feedback. COLA consists of three specialized agents—a Classifier that narrows large schemas to relevant sub-schemas, an Analyzer that plans and produces partial SQL, and a Corrector that validates and revises the final query using execution results. Integrated with strong LLM backbones such as OpenAI-o1 and DeepSeek-R1, COLA raises execution accuracy on the challenging MultiSpider 2.0 benchmark from ~4 % (single-LLM reasoning) to 13-16 % without any task-specific fine-tuning.

267199832.c3

Supervised warm-up of policy for structured reasoning RL

SEER: Facilitating Structured Reasoning and Explanation via Reinforcement Learning

SEER first converts the structured reasoning task (entailment trees or reasoning graphs) into a collection of single-step supervised examples and fine-tunes a T5-large policy network on this data. This warm-up yields a stable initial policy, mitigates sample inefficiency, and accelerates convergence when the subsequent PPO-based reinforcement-learning phase begins.

267199832.c2

Fine-grained reward function distinguishing correct, erroneous, and redundant steps

SEER: Facilitating Structured Reasoning and Explanation via Reinforcement Learning

The authors introduce a reward scheme for structured reasoning in which each reasoning step receives a scalar reward: +1 for steps that perfectly match a gold intermediate node, -0.5 for redundant steps that do not contribute to the final entailment tree, and -1 for erroneous steps. This granularity provides informative feedback to the reinforcement-learning policy, encouraging exploration of useful actions while penalizing harmful moves, and improves training stability and the quality of generated structured explanations.

271921406.c6

Comprehensive Empirical Evaluation of FRVA on EntailmentBank

FRVA: Fact-Retrieval and Verification Augmented Entailment Tree Generation for Explainable Question Answering

The paper conducts extensive experiments on the three EntailmentBank tasks (no-distractor, distractor, full-corpus) comparing the proposed FRVA system against direct-generation and stepwise baselines. It reports quantitative improvements in leaf-node F1, proof-step correctness, intermediate-conclusion accuracy, and overall tree accuracy. Detailed ablation studies isolate the impact of fact retrieval, bidirectional generation, cross-verification, and multi-view contrastive learning, and an error analysis characterises common failure modes.

268247795.c0

xSTREET multilingual structured-reasoning benchmark

Eliciting Better Multilingual Structured Reasoning from LLMs through Code

The authors extend the English STREET benchmark to six languages (English, Arabic, Spanish, Russian, Chinese, Japanese) by translating all natural-language statements while preserving the original structured reasoning graphs. They create a high-quality test split that is manually post-edited by expert translators and provide machine-translated training and development splits. The benchmark comprises four complex reasoning tasks (ARC, GSM8K, AQUA_RAT, AR_LSAT) with defined answer formats, accuracy-based evaluation, and publicly released code and data, enabling systematic assessment of multilingual structured reasoning in LLMs.

267199832.c1

Structure-based return for tree and graph reasoning

SEER: Facilitating Structured Reasoning and Explanation via Reinforcement Learning

The paper introduces a novel structure-based return for reinforcement-learning agents that perform structured reasoning over entailment trees or reasoning graphs. Instead of the conventional chained cumulative reward, the return for a state is defined as the immediate reward plus the discounted average value of its parent nodes in the reasoning structure, thereby respecting hierarchical and branching dependencies. This formulation ensures that equivalent reasoning trajectories receive identical returns, addresses instability in prior methods, The authors analyze how this return resolves the equivalent-trajectory problem and improves training stability and effectiveness.

280322203.c3

State-of-the-Art Empirical Results on EntailmentBank

Natural Logic at the Core: Dynamic Rewards for Entailment Tree Generation

The NLDR system is evaluated on the EntailmentBank benchmark across three increasingly challenging tasks (no-distractor, distractor, full-corpus). It achieves 42.6 % on Task 1, 36.5 % on Task 2, and 13.2 % on Task 3, surpassing strong baselines such as SEER, RLET, METGEN, and GPT-4 variants with Chain-of-Thought or Tree-of-Thought prompting. These results demonstrate that dynamic, natural-logic-driven rewards improve multi-step entailment-tree generation over prior static-reward or heuristic methods.

267199832.c5

Open-source release of SEER code and reproducibility package

SEER: Facilitating Structured Reasoning and Explanation via Reinforcement Learning

The authors publicly release the full implementation of the SEER framework, including the policy network, critic network, reward computation modules, training scripts, and data-preprocessing utilities at https://github.com/Chen-GX/SEER. The package contains environment specifications, hyper-parameter settings, and evaluation scripts to enable exact replication of the reported results on structured reasoning benchmarks and to facilitate extension to new tasks.

268863949.c3

Empirical evaluation of SPEH on EntailmentBank Task 1

Step Feasibility-Aware and Error-Correctable Entailment Tree Generation

The authors conduct a comprehensive empirical study on the EntailmentBank Task 1 test split, comparing the proposed SPEH framework (with Step Feasibility Perception and State Error Handling enabled) against strong baselines including EntailmentWriter, IRGR, RLET, METGEN, and ChatGPT-3.5. SPEH achieves the highest Overall AllCorrect score (37.7 %) and shows improvements in step and intermediate metrics, demonstrating that the introduced mechanisms effectively enhance entailment-tree generation performance.

268358481.c6-1

State-of-the-art empirical performance of LMPM on EntailmentBank

A Logical Pattern Memory Pre-trained Model for Entailment Tree Generation

The paper evaluates the LMPM model, integrated into the METGEN framework, on the EntailmentBank benchmark (tasks 1-3) and shows that METGEN+LMPM outperforms prior systems (EntailmentWriter, IRGR, RLET, METGEN+T5) on automatic metrics (Leaves, Steps, Intermediates F1 and AllCorrect) and on human judgments of validity, logical consistency, readability, and reasonability.

280322203.c1

NLDR: Natural-Logic-Based Dynamic-Reward Framework for Entailment-Tree Generation

Natural Logic at the Core: Dynamic Rewards for Entailment Tree Generation

NLDR is a reinforcement-learning framework that integrates natural-logic entailment scores as step-wise dynamic rewards for constructing entailment trees. It employs a two-state deterministic finite automaton (DFA) to aggregate predicted natural-logic relations, producing a reward that reflects global logical consistency rather than a fixed scalar. The framework couples this reward module with a T5-large policy model that jointly selects premises and generates intermediate conclusions, yielding an end-to-end system for multi-step structured reasoning.

280322203.c2

Joint Premise Selection and Intermediate Conclusion Generation in RL Policy

Natural Logic at the Core: Dynamic Rewards for Entailment Tree Generation

The paper extends the reinforcement-learning (RL) policy for entailment-tree construction to output a combined action consisting of (i) a set of selected premises and (ii) a generated intermediate textual conclusion at each reasoning step. This joint action space enables the reward function to evaluate both premise relevance and logical soundness of the generated inference, allowing the agent to learn coherent multi-step reasoning rather than isolated premise-pair decisions. The policy is instantiated as a T5-large generative model and optimized with proximal policy optimization (PPO) to maximize the dynamic natural-logic-based rewards.

267199832.c0

SEER: General RL Framework for Structured Reasoning and Explanation

SEER: Facilitating Structured Reasoning and Explanation via Reinforcement Learning

SEER is a reinforcement-learning-based system that formulates the generation of structured explanations (entailment trees or reasoning graphs) for question answering as a sequential decision-making problem. It integrates a generative policy capable of selecting arbitrary premise combinations, a structure-based return that respects hierarchical and branching dependencies, and a fine-grained reward that differentiates correct, erroneous, and redundant steps. The framework is trained via a supervised warm-up followed by PPO-based RL fine-tuning and achieves state-of-the-art performance on multiple structured reasoning benchmarks.

271921406.c0

FRVA: Integrated Fact Retrieval and Bidirectional Proof Generation Framework for Explainable QA

FRVA: Fact-Retrieval and Verification Augmented Entailment Tree Generation for Explainable Question Answering

FRVA is a complete system for generating structured entailment trees that explain question-answer pairs. It first filters irrelevant knowledge sentences with a neural fact-retrieval module, then constructs proof steps using two complementary seq2seq generators: a deductive (bottom-up) and an abductive (top-down) generator. Generated steps are cross-validated by a verifier and further refined with a multi-view contrastive learning objective that aligns facts and generated steps with the hypothesis. Finally, the deductive and abductive trees are aligned and fused into a single optimal entailment tree, reducing combinatorial search space, mitigating error propagation, and achieving state-of-the-art performance on the EntailmentBank benchmark.

253244813.c0

Premise Recovery Task for Natural Language Deduction with Incomplete Information

Natural Language Deduction with Incomplete Information

The paper defines a new reasoning task in which a system must construct a natural-language proof (entailment tree) that both entails a given goal and recovers a missing premise that has been deliberately omitted from the initial set of premises. This formulation extends prior entailment-tree tasks by requiring abductive inference to materialize unstated knowledge, and introduces evaluation criteria for premise coverage and step validity. The task is motivated by the need for explainable reasoning systems that can handle underspecified information while producing verifiable intermediate steps.

258959919.c1

Algorithmic pipeline for transformer-guided chaining based on forward chaining

Investigating Transformer-guided Chaining for Interpretable Natural Logic Reasoning

The paper formalizes a neuro-symbolic inference pipeline that adapts the classic forward-chaining algorithm to operate with pretrained transformer models. At each iteration a T5 encoder-decoder model performs rule matching and generates new facts (rule inference), the accumulated fact set is expanded, and a RoBERTa classifier checks whether the hypothesis is entailed, contradicted, or still unknown. The loop continues until no new facts are produced or the hypothesis is resolved, and the sequence of applied rules and generated facts is assembled into an interpretable reasoning chain. Pseudocode and detailed procedural steps are provided to ensure reproducibility.

258343907.c0

Deductive-Additivity-Based Planning Heuristic and Proof-Generation Search Algorithm

Deductive Additivity for Planning of Natural Language Proofs

The paper proposes a planning heuristic that ranks candidate premise pairs by the cosine similarity between the summed embeddings of the pair and the embedding of the goal statement. This heuristic is embedded in a full proof-generation search algorithm (Algorithm 1) that iteratively expands an entailment tree: it caches embeddings of the initial fact pool, computes summed premise vectors, selects the highest-scoring pair, generates a deduction with a language model, validates the deduction with an entailment classifier, and repeats until the claim is proved or a step limit is reached. The approach enables rapid, scalable multi-step reasoning without expensive transformer-based pairwise scoring at each step.

253244813.c2

ADGV: Bidirectional Search over Abductive and Deductive Generation with Validation

Natural Language Deduction with Incomplete Information

The authors introduce ADGV, a search-based reasoning system that interleaves forward-chaining deductive steps and backward-chaining abductive steps using two priority-queue fringes. At each iteration the system scores candidate generations from a deductive step model (S^d) and an abductive step model (S^a), filters them with round-trip consistency validators, and pushes validated results back onto the fringes. This bidirectional heuristic search enables discovery of a missing premise while simultaneously constructing a complete entailment tree that proves the goal.

253244813.c1

Everyday Norms: Why Not? (ENWN) Dataset

Natural Language Deduction with Incomplete Information

The authors create a new English-language resource of 100 manually authored entailment trees that capture everyday ethical scenarios. Each tree includes a set of premise statements, intermediate deductive steps, a goal statement, and a deliberately omitted gold premise. The dataset enables evaluation of systems that must both recover the missing premise and construct a proof under incomplete information. The dataset and accompanying code are released publicly, providing a benchmark for abductive reasoning and commonsense normative knowledge in natural language deduction.

249062748.c0

NLProofS: Verifier-Guided Stepwise Proof Generation

Generating Natural Language Proofs with Verifier-Guided Search

NLProofS is a stepwise natural-language proof generation framework that conditions each generated proof step on the hypothesis, employs an independently trained verifier to score the logical validity of candidate steps, and conducts a global search over a directed-acyclic proof graph to select a proof tree with maximum aggregated verifier scores. The prover (a fine-tuned T5 model) generates multiple candidate steps per iteration, the verifier (a fine-tuned RoBERTa model) assigns a validity score to each step, and a min-aggregation scoring function propagates scores through the graph. A beam-search-based proof-generation algorithm iteratively updates the graph until convergence, improving the efficiency and accuracy of the hypothesis and contain fewer hallucinated steps.

253244813.c4

Round-Trip Validation Mechanisms for Logical Consistency

Natural Language Deduction with Incomplete Information

The paper introduces a suite of validation modules that enforce bidirectional consistency between forward (deductive) and backward (abductive) generation steps during proof search. Deductive Agreement checks that an abductively generated hypothesis, when combined with its premise, reproduces the original conclusion via the deductive model. Abductive Agreement verifies that each premise of a deductive step can be recovered by an abductive generation conditioned on the conclusion. Additional filters (de-duplication and consanguinity pruning) remove redundant or unsafe generations before they are added to the search frontier, thereby improving the soundness of the constructed entailment trees.

253244813.c1

Empirical Evaluation and Error Analysis of ADGV

Natural Language Deduction with Incomplete Information

The authors conduct a comprehensive empirical study of the ADGV (Abduction and Deductive Generation with Validation) system on two benchmark suites: the EntailmentBank test set and the newly introduced Everyday Norms: Why Not? (ENWN) dataset. They compare ADGV against four baselines—deductive-only (DG), abductive-only (AG), combined deductive-abductive without validation (ADG), and an end-to-end T5-3B model (E2E)—using three quantitative metrics: premise-coverage (recovery scoring), human-annotated step validity, and deductive-agreement scores derived from round-trip validation. Additionally, they perform a qualitative error analysis that isolates systematic failure modes such as underspecification, cascading errors, and premise subsumption, and discuss directions for future improvements.

258840551.c1

Reasoning Graph Formalism for Structured Natural-Language Explanations

STREET: A Multi-Task Structured Reasoning and Explanation Benchmark

The paper defines a reasoning graph as a directed acyclic graph whose vertices are textual logical units (TLUs) extracted from a question, context, and answer options, and whose edges connect premises to intermediate conclusions, terminating at an answer node. This formulation captures multi-premise, multi-step reasoning in natural language while remaining machine-readable, extending prior entailment-tree ideas by allowing arbitrary DAG structures and by integrating the answer directly into the graph.

253257103.c0

MDP-Formulated Reinforcement Learning Framework for Entailment-Tree Generation (RLET)

RLET: A Reinforcement Learning Based Approach for Explainable QA with Entailment Trees

The paper introduces RLET, a framework that formulates the construction of entailment trees for explainable question answering as a Markov Decision Process (MDP). At each step the state consists of the set of unused fact and intermediate sentences, the action selects a pair of sentences, and a deterministic transition adds the newly generated intermediate conclusion. A cumulative, alignment-based reward is used to train a stochastic sentence-selection policy with policy-gradient reinforcement learning, while a separate deduction-generation module (rule-based or Seq2Seq) produces the intermediate conclusions. The architecture is instantiated with a DeBERTa-v3-base encoder for selection and a BART-large (or T5-based) generator for deduction.

273641238.c1

Definition of Multimodal Entailment-Tree Generation Task

An Entailment Tree Generation Approach for Multimodal Multi-Hop Question Answering with Mixture-of-Experts and Iterative Feedback Mechanism

The paper formally defines a multimodal entailment-tree generation task for multi-hop question answering. Given a set of premises (facts extracted from text, tables, and images) and a hypothesis (the original question), the task requires generating a tree whose leaf nodes are selected premises and whose internal nodes are intermediate conclusions produced by a large language model. Unlike prior entailment-tree tasks in pure NLP, the answer is not provided during training, making the formulation applicable to multimodal settings where the answer must be inferred after tree construction.

269804127.c0

State-aware iterative fact retrieval for multi-step Tree-structured Entailment Reasoning (TER)

Empowering Tree-structured Entailment Reasoning: Rhetorical Perception and LLM-driven Interpretability

The paper introduces a dynamic fact-retrieval mechanism for multi-step TER that trains the fact-selection controller iteratively across reasoning steps. At each step the controller conditions on a mean-pooled CLS embedding of the current reasoning state and the hypothesis, and selects "good" facts defined by lexical overlap with the hypothesis or intermediate conclusion and structural proximity to the newly built node. The training objective combines a max-margin ranking loss for positive/negative fact discrimination with a negative log-likelihood term for optimal-fact selection, enabling on-the-fly premise retrieval and reducing error propagation, which yields higher leaf-fact accuracy on EntailmentBank distractor and full-corpus settings.

280322203.c1

Natural-Logic-Guided Reward Pipeline for Entailment-Tree Generation

Natural Logic at the Core: Dynamic Rewards for Entailment Tree Generation

The paper introduces a multi-stage pipeline that converts natural-logic reasoning into step-wise dynamic rewards for entailment-tree construction. First, an instruction-tuned LLM (e.g., Llama 3.1-8B-Instruct) semantically chunks the hypothesis and candidate facts. Second, the Hungarian algorithm aligns chunks to obtain optimal span correspondences. Third, a fine-tuned DeBERTa model predicts the natural-logic relation (equivalence, forward entailment, etc.) for each aligned pair. Fourth, a monotonicity classifier projects these relations according to contextual upward/downward monotonicity. Finally, the projected relation sequence is processed by a two-state deterministic finite automaton whose output probability forms the dynamic reward used in reinforcement learning, providing fine-grained logical feedback at every reasoning step.

281421016.c0

Integrative Grounding Task Definition

INTEGROUND: On the Evaluation of Verification and Retrieval Planning in Integrative Grounding

The paper formally defines the integrative grounding task, wherein a language model must retrieve and verify multiple inter-dependent pieces of external evidence to support a hypothesis proposition. This extends traditional grounding, which assumes a single supporting document, by requiring synthesis of several evidential snippets to answer complex real-world queries. The definition introduces formal notions of propositions, knowledge bases, and the grounding objective (find a subset Σ ⊆ K such that Σ entails or contradicts the hypothesis).

258480186.c7-1

Multi-hop reasoning identified as a primary challenge for clinical-trial NLI

SemEval-2023 Task 7: Multi-Evidence Natural Language Inference for Clinical Trial Data

The authors observe that successful inference on the NLI4CT benchmark frequently requires combining information from multiple sentences of a clinical trial report, i.e., multi-hop reasoning. Empirical analysis of system performance shows that smaller or purely discriminative models struggle disproportionately on instances requiring longer inference chains, highlighting multi-hop reasoning as a key obstacle.

258479664.c0

Problem formulation and modular framework for Faithful QA (FAME)

Faithful Question Answering with Monte-Carlo Planning

The paper reconceives the Faithful Question Answering (FQA) task as a discrete decision-making problem in which a reasoning environment maintains a state consisting of a hypothesis, a partial entailment tree, and a set of candidate premises. A generative controller (implemented with a T5-large model) observes this state and proposes actions—Retrieve, Entail, or End—to iteratively construct an entailment tree whose intermediate steps are valid and whose root conclusion faithfully supports the answer. The modular architecture (retriever, single-step entailment module, verifier) enables explicit look-ahead planning over reasoning actions rather than greedy step-wise generation.

258479664.c1

Monte-Carlo planning algorithm for foresighted action selection

Faithful Question Answering with Monte-Carlo Planning

The paper adapts Monte-Carlo Tree Search (MCTS) with an Upper Confidence Bound (UCB) to the discrete decision-making problem of constructing entailment trees for question answering. In each simulation the planner selects an action (Retrieve, Entail, End) based on a combination of the controller's prior probability and a state-verifier score, updates the action value Q for the selected action, and back-propagates the resulting state score. This look-ahead search yields actions that are expected to lead to high-quality, valid, and faithful reasoning trees, outperforming greedy, over-generate-and-filter, and beam-search baselines.

252781096.c0

TemplateCSR task formulation

Template Filling for Controllable Commonsense Reasoning

The paper defines TemplateCSR, a new commonsense reasoning task that frames reasoning as filling a natural-language template with open-vocabulary concept slots, a multiple-choice qualifier slot, and an optional free-form explanation slot. A language model receives the template together with natural-language instructions describing each slot and must generate a fully-filled sentence that respects these constraints. This formulation departs from traditional multiple-choice QA and open-domain knowledge-base retrieval settings, providing fine-grained controllability over the reasoning chain. The task is presented as a sequence-to-sequence generation problem suitable for instruction-fine-tuned models.

258343907.c0

Deductive Additivity Property for Sentence Embeddings

Deductive Additivity for Planning of Natural Language Proofs

The paper defines deductive additivity (DA) as a geometric property of a sentence-embedding space: for any two premise sentences p_a and p_b that entail a conclusion d_{ab}, the vector sum of their embeddings E(p_a)+E(p_b) must have higher cosine similarity to the embedding of the conclusion E(d_{ab}) than to the embedding of any non-entailed statement x. This formal inequality cos(E(p_a)+E(p_b),E(d_{ab})) > cos(E(p_a)+E(p_b),E(x)) reframes the planning stage of natural-language proof generation as a vector-arithmetic search problem, enabling fast, scalable selection of reasoning steps via simple addition and cosine similarity.

246015349.c0

SCSEARCH: Factored system for natural-language deduction

Natural Language Deduction through Search over Statement Compositions

The paper introduces SCSEARCH, a modular architecture that performs natural-language deduction by (i) a step-deduction model that generates a direct conclusion from a pair of premises; (ii) a best-first search procedure that explores compositions of statements guided by a learned heuristic, and (iii) a goal-entailment model that decides when the hypothesis has been proved. By factorising generation, search, and verification, the system guarantees that each intermediate step is soundly derived from its inputs, yielding an explicit entailment tree rather than a black-box generation. SCSEARCH is implemented with pretrained T5-3B for step generation and DeBERTa-large for goal checking, and can handle arbitrary numbers of premises and binary-branching trees.

248887255.c3

Linearized entailment-tree representation for seq2seq generation

Entailment Tree Explanations via Iterative Retrieval-Generation Reasoner

The paper proposes a plain-text linearization scheme that encodes an entailment tree as a flat token sequence suitable for a standard sequence-to-sequence model. Each leaf sentence is prefixed with the token "sent", each generated intermediate conclusion with "int", and the hypothesis with "hypothesis"; logical relations are expressed with "&" for conjunction and "->" for entailment. This representation enables the T5 generator to produce one entailment step at a time while preserving the hierarchical structure required for downstream alignment and evaluation.

261822669.c0-1

Compositional Generalization Formulation for Structured Explanation Generation

Dynamic MOdularized Reasoning for Compositional Structured Explanation Generation

The paper reconceptualizes the Structured Explanation Generation (SEG) task to explicitly evaluate compositional reasoning by introducing two orthogonal dimensions: productivity, which measures the ability to handle longer proof lengths than seen during training, and systematicity, which measures the ability to recombine known reasoning components into novel proof shapes. This formulation builds on the established SEG task and adapts the compositional generalization framework from prior work to the sentence-level entailment-tree setting.

261822669.c0-2

Compositional Benchmark Splits for Structured Explanation Generation (EntB-Length, EntB-Shape, DBP-Length)

Dynamic MOdularized Reasoning for Compositional Structured Explanation Generation

The authors reorganize two existing resources—EntailmentBank and a synthetic DBpedia rule-based dataset—into three new benchmark splits that each isolate a single compositional challenge. EntB-Length tests productivity by varying proof length, EntB-Shape tests systematicity by varying proof shape, and DBP-Length tests productivity on a rule-based dataset. They provide detailed statistics and release the split files for community use, enabling systematic evaluation of compositional reasoning in structured explanation generation.

261822669.c0-3

Procedure for Constructing Compositional SEG Evaluation Splits

Dynamic MOdularized Reasoning for Compositional Structured Explanation Generation

The paper presents a reproducible protocol for partitioning existing Structured Explanation Generation (SEG) datasets into compositional evaluation splits. The procedure extracts the proof length of each entailment tree, classifies trees according to predefined shape templates, assigns primitive-level instances (short proofs or simple shapes) to the training set and compositional-level instances (longer proofs or composed shapes) to the test set, and provides scripts together with split statistics to ensure consistent generation for future research.

267782534.c0

Decompositional RTE task definition

Enhancing Systematic Decompositional Natural Language Inference Using Informal Logic

The paper formally defines the Decompositional Recognizing Textual Entailment (RDTE) task: given a question Q, a hypothesis h, and a set of premises generated as a decomposition of h, determine whether the premises are contextually sufficient to prove h. This extends traditional single-premise NLI to recursive, multi-premise reasoning required by entailment-tree search algorithms and provides a clear protocol for labeling compositional entailments.

268091324.c0

Multimodal Entailment Tree Generation Task

TV-TREES: Multimodal Entailment Trees for Neuro-Symbolic Video Reasoning

The paper defines a new task that requires generating entailment trees from both video frames and dialogue transcripts to prove question-answer pairs in video-language QA. The task extends prior text-only entailment-tree generation by specifying a multimodal evidence bank (V ∪ D), a recursive tree structure where each node's evidence must entail its hypothesis, and evaluation criteria of acceptability, relevance, and sufficiency. This formulation provides a systematic, interpretable reasoning benchmark for complex TV clips and enables future multimodal neuro-symbolic systems to be evaluated on structured proof generation.

268091324.c1-1

Recursive Multimodal Entailment Tree Generation Algorithm

TV-TREES: Multimodal Entailment Trees for Neuro-Symbolic Video Reasoning

A novel recursive search algorithm that constructs multimodal entailment trees by iteratively retrieving candidate evidence, filtering premises, and decomposing hypotheses into sub-hypotheses. The algorithm operates over a joint evidence bank of video frames and dialogue transcripts, producing a proof tree whose leaves are grounded in concrete multimodal evidence and whose internal nodes represent logical entailment steps. It orchestrates retrieval, NLI-based filtering, LLM-driven decomposition, and vision-language model verification within a unified neuro-symbolic framework.

266177015.c0

HTL dataset: fine-grained natural-language Hoare triples for program semantics

Natural Language Annotations for Reasoning about Program Semantics

The paper introduces HTL (Hoare Triples via Language), a publicly released dataset that pairs self-contained Python code snippets with natural-language pre- and post-conditions expressed as Hoare triples. Annotations are collected at the granularity of individual statements and small nested spans. The guidelines enforce logical correctness, compositionality, and a uniform tokenisation of variable identifiers, and are supported by statistical analyses of annotation length and coverage. This product underpins the creation of a coherent, machine-readable corpus of Hoare-style triples expressed in natural language.

266177015.c1

Annotation protocol and guidelines for natural-language Hoare triples

Natural Language Annotations for Reasoning about Program Semantics

The authors devise a systematic protocol that directs annotators to write pre- and post-conditions in natural language that are immediate logical entailments of a highlighted code span and that compose consistently across nested spans. The guidelines enforce logical correctness, compositionality, and a uniform tokenisation of variable identifiers, and are supported by statistical analyses of annotation length and coverage. This protocol underpins the creation of a coherent, machine-readable corpus of Hoare-style triples expressed in natural language.

266177015.c4-2

Position statement: NL annotations enable explainable program assistants with Hoare-style entailment trees

Natural Language Annotations for Reasoning about Program Semantics

The authors argue that natural-language pre- and post-conditions annotated for code statements can be leveraged by language models to build step-wise entailment trees analogous to Hoare triples, providing transparent explanations of program execution and supporting both deductive and abductive reasoning about code behavior.

265150086.c0

ConDec: Contrastive Learning-Based Stepwise Decoding for Natural Language Proof Generation

Are LLMs Rigorous Logical Reasoners? Empowering Natural Language Proof Generation by Stepwise Decoding with Contrastive Learning

ConDec introduces a training regime that couples stepwise proof generation with a contrastive learning objective. During fine-tuning, the model maximises the similarity between the encoder representation of the current proof context and the decoder representation of the correct next step, while minimising similarity to specially crafted hard-negative steps that mimic common decoding errors (repetition and invalid entailment). The approach is applied to an encoder-decoder LLM (Flan-T5-Large) and operates as a single-stage generator, removing the need for external search or verification components.

267750612.c0-1

Analysis of missing backtracking in Least-to-most prompting

SymBa: Symbolic Backward Chaining for Structured Natural Language Reasoning

The paper conducts a systematic examination of the Least-to-most prompting method, demonstrating that it omits the backtracking step of the classic SLD Resolution algorithm. Minimal counter-examples are provided to show how the lack of backtracking leads to failed proof construction and reduced answer accuracy on structured reasoning benchmarks. By contrasting the implementation with the complete SLD procedure, the analysis reveals a theoretical incompleteness in the method.

267750612.c0-2

Analysis of missing binding propagation in LAMBADA

SymBa: Symbolic Backward Chaining for Structured Natural Language Reasoning

The paper examines the LAMBADA backward-chaining system and shows that it implements binding propagation only from goal to subgoals, omitting propagation between subgoals and from subgoals back to the goal. Counter-examples illustrate how this limitation prevents correct reasoning in relational and arithmetic tasks, leading to incompleteness compared with the full SLD Resolution algorithm.

268358481.c0

Logical Pattern Memory Pre-trained Model (LMPM) architecture

A Logical Pattern Memory Pre-trained Model for Entailment Tree Generation

The paper introduces LMPM, an entailment-tree generation model that augments a T5 encoder-decoder with an external memory module storing latent representations of logical deduction patterns. Premises are encoded, an address network (using Gumbel-softmax for single-pattern selection or softmax for multi-pattern combination) retrieves a pattern vector from the memory, and this vector is added to the decoder start token to guide generation of logically consistent intermediate conclusions. This design separates logical pattern learning from language modeling, enabling more reliable multi-step reasoning.

270257656.c0

ACCORD conceptual framework for anti-factual commonsense reasoning

***ACCORD**: Closing the Commonsense Measurability Gap*

The paper introduces ACCORD, a formally defined framework that augments existing commonsense QA datasets with anti-factual contexts. ACCORD specifies a set of reasoning skills, pairing templates, and recursively reducible reasoning trees (polytrees), enabling precise control over the number of reasoning hops and distractors. Variables in the generated trees are grounded anti-factually via a back-tracking search that ensures the answer depends on the context rather than prior ConceptNet, ensuring that models cannot rely on memorized knowledge. The framework is positioned as a means to close the measurability gap between commonsense and formal reasoning by providing construct-valid, scalable benchmarks.

270799239.c0

Definition of the Compositional Structured Explanation Generation (CSEG) task

Compositional Structured Explanation Generation with Dynamic Modularized Reasoning

The paper introduces a new task, Compositional Structured Explanation Generation (CSEG), which requires a model to generate a multi-step entailment tree (structured explanation) for a hypothesis given a set of candidate sentences. CSEG explicitly evaluates compositional generalization by testing whether models can extrapolate from trees with a limited number of inference steps to trees with more steps, focusing on both proof length (productivity) and tree shape (systematicity). The task is framed as a generation problem, combining reasoning and compositionality challenges in a single end-to-end setting.

270799239.c1

Construction of CSEG benchmark splits (EntailmentBank-Length, EntailmentBank-Shape, DBpedia-Length)

Compositional Structured Explanation Generation with Dynamic Modularized Reasoning

The authors curate three compositional generalization splits from existing corpora to evaluate the proposed CSEG task. EntailmentBank-Length (EntB-L) groups entailment trees by proof length to test productivity, while EntailmentBank-Shape (EntB-Sh) groups them by tree shape to test systematicity. DBpedia-Length (DBP-L) is a synthetic rule-based dataset derived from DBpedia, also partitioned by proof length. These splits provide standardized evaluation protocols for length and shape generalization in structured explanation generation.

273641238.c0

Joint Entailment-Tree Generation and Answering Formulation for Multimodal Multi-Hop QA

An Entailment Tree Generation Approach for Multimodal Multi-Hop Question Answering with Mixture-of-Experts and Iterative Feedback Mechanism

The paper reconceives multimodal multi-hop question answering (MMQA) as a joint problem of (i) generating an entailment tree that explicitly structures the logical relationships among multimodal evidence (facts) and (ii) producing the final answer conditioned on that tree. By treating the original question as a hypothesis and a constructed fact base as premises, the formulation forces the system to select only relevant evidence, model intermediate reasoning steps, and provide interpretable explanations, thereby addressing redundancy and opacity of prior text-conversion pipelines.

273641238.c5-3

Ablation Study Quantifying Component Contributions

An Entailment Tree Generation Approach for Multimodal Multi-Hop Question Answering with Mixture-of-Experts and Iterative Feedback Mechanism

The paper conducts systematic ablation experiments that individually remove or replace four major modules—question decomposition with sub-question image captioning, the multi-task mixture-of-experts (MoE) component, the iterative feedback loop, and the explicit entailment-tree generation step—to measure each module's impact on final multimodal multi-hop QA accuracy. Results show measurable gains from every component, confirming their necessity for state-of-the-art performance.

273323372.c1

Graph-based structure-aware demonstration selection

Exploring the Role of Reasoning Structures for Constructing Proofs in Multi-Step Natural Language Reasoning with Large Language Models

The paper proposes a method for selecting few-shot exemplars for in-context learning by encoding the proof graphs of candidate demonstrations with a Graph Attention Network (GATv2) and computing cosine similarity over the resulting embeddings. For a target problem, the system retrieves the most similar demonstrations from a proof database, allowing the language model to observe proof-structure patterns that are directly relevant to the current reasoning task. This replaces prior fixed-demo or purely text-similar demonstration selection approaches.

273323372.c3

Proof-hint generation module

Exploring the Role of Reasoning Structures for Constructing Proofs in Multi-Step Natural Language Reasoning with Large Language Models

The paper introduces a dedicated proof-hint generation component that receives an intermediate conclusion produced by the LLM, compares it with the target hypothesis, and outputs a concise textual hint describing the missing reasoning step. This hint is fed back into the candidate-retrieval module to guide the selection of evidence sentences for the next reasoning step, improving the efficiency and accuracy of multi-step proof construction within the overall framework.

273162523.c0

MDP formulation of multi-step reasoning using entailment graphs

Deliberate Reasoning in Language Models as Structure-Aware Planning with an Accurate World Model

The paper formalizes multi-step reasoning as a Markov Decision Process (MDP) where each reasoning state is an entailment graph that encodes statements and their logical dependencies. Actions correspond to graph-expansion steps that add new statements or entailment edges, and the transition function is modeled by a learned world model that predicts the updated graph. This formulation reframes reasoning as sequential planning over structured knowledge, enabling symbolic verification of intermediate steps and providing a principled reward-free decision mechanism via a discriminator.

273162523.c1

SWAP (Structure-aware Planning with an Accurate World Model) framework

Deliberate Reasoning in Language Models as Structure-Aware Planning with an Accurate World Model

SWAP is an end-to-end reasoning system that treats multi-step problem solving as planning over entailment graphs. It combines a policy model that proposes candidate actions, an accurate world model that predicts graph updates, a discriminator that re-ranks multiple candidates using contrastive ranking, and a controller that decides when to stop. The framework is instantiated by fine-tuning a single large language model (via LoRA) to serve all three roles and incorporates diversity-based modeling to explore a broader solution space. By explicitly constructing and verifying entailment graphs, SWAP enables symbolic verification of intermediate steps and improves consistency on math, logic, and coding benchmarks.

273162523.c2

Entailment-graph representation for reasoning states

Deliberate Reasoning in Language Models as Structure-Aware Planning with an Accurate World Model

The paper encodes each reasoning state as a directed (hyper)graph whose nodes are natural-language statements (evidence, assumptions, lemmas, conclusions) and whose (hyper)edges represent entailment relations mapping a set of premises to a conclusion. This graph-based encoding enables symbolic verification of intermediate steps, tracks dependencies and inconsistencies, and serves as the world state for multi-step planning. By treating the graph as the state, the system can select actions that expand or modify the graph in a principled way.

248887255.c1

Conditional dense-retrieval algorithm for premise selection

Entailment Tree Explanations via Iterative Retrieval-Generation Reasoner

The paper introduces a conditional dense-retrieval procedure (Algorithm 1) that selects k premises in two phases: an initial set retrieved using only the hypothesis as query, and a second set retrieved with a query that concatenates the hypothesis and previously selected premises. A conditioning factor q determines how many premises are taken from each phase, allowing the system to prioritize lexically close leaf nodes while still gathering deeper, less-similar premises needed for later proof steps. The algorithm defines a specific query representation for dense passage retrieval and is integrated into the iterative IRGR framework, yielding higher premise recall and improved entailment-tree explanation performance.

233297051.c0

Formulation of explanation as multistep entailment trees

Explaining Answers with Entailment Trees

The paper reconceives open-domain textual question-answering explanation as the construction of a directed entailment tree, where each node is a multi-premise textual entailment step that incrementally derives the hypothesis (question + answer) from known facts. This formulation separates the correctness of the derivation from its utility for users, enabling systematic evaluation of explanations independent of answer selection. By defining explanations as trees rather than isolated rationales, the work provides a high-level conceptual framework that can guide future research on explainable QA.

Prerequisite Edge Description

Description: Techniques for modeling logical entailment relations between natural-language statements, used to build explanatory reasoning chains such as entailment trees.

Explanation: The claim relies on treating NL pre/post-conditions as entailments that can be composed analogously to Hoare triples.

strength: strong; depth: 2

Prerequisite Edge Description

Description: Empirical observation that leaf facts often share non-stop-word overlap with the hypothesis or intermediate conclusions, used to label positive retrieval candidates.

Explanation: Defining "good" facts via lexical overlap is a key component of the retrieval training objective.

strength: strong; depth: 1